Fiscal Year:	FY 2021	**Task Last Updated:**	FY 10/01/2021
PI Name:	Nicholson, Wayne Ph.D.		
Project Title:	Global Transcriptome Profiling to Identify Cellular Stress Mechanisms Responsible for Spaceflight-Induced Antibiotic Resistance		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	32953	**Congressional District:**	3
Comments:	NOTE: Congressional District is for University of Florida (32611-0001)--Ed., 8/17/2012; changed to 3 (Ed., 7/30/13)		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	10/01/2014	**End Date:**	09/30/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 9/30/2021 per NSSC information (Ed., 9/10/20) NOTE: Extended to 9/30/2020 per NSSC information (Ed., 10/24/2019) NOTE: Extended to 9/30/2019 per NSSC information (Ed., 1/11/2019) NOTE: Extended to 9/30/2018 per NSSC information (Ed., 10/11/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Fajardo-Cavazos, Patricia Ph.D. (University of Florida)		
Grant/Contract No.:	NNX14AT38G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Responses of various microorganisms to the spaceflight environment have been examined by numerous studies conducted either in space or on the ground in clinostats. The results clearly show that several microbial behaviors differ in space, but not in a consistent way. The underlying mechanism(s) involved are not explainable by a single model. Exposure to the spaceflight environment (e.g., microgravity, radiation) alters a microbial cell's immediate surroundings, lowering convective mass and heat transfer or reducing mechanical shear forces. Changes in such fundamental physical forces affect the rates at which gases, nutrients, signaling molecules, and waste products are exchanged between microbes and their surroundings. Microbes perceive these alterations as environmental stress (i.e., the "spaceflight syndrome") and mount a complex set of stress responses. One consequence of the spaceflight stress response is alteration in their susceptibility to antibiotics. This could be due to mechanisms such as: production of biofilms, alteration of cell surfaces, up-regulation of drug efflux systems, or an increased rate of mutation in genes encoding drug targets. All of these mechanisms have been well documented as outputs of global microbial stress responses. Of particular interest is oxidative stress, which has been shown both to trigger maladaptive responses in higher organisms including astronauts, and to increase bacterial resistance to antibiotics. Central Hypothesis: Microorganisms subjected to the integrated spaceflight environment invoke a spectrum of stress responses, some leading to alterations in their antibiotic susceptibility. The underlying mechanisms can be identified using transcriptome profiling of model organisms. Specific Aims: We postulate that spaceflight stress responses vary depending on the organism studied because each species of microbe is endowed with its own unique collection of stress response systems. In order to probe the underlying mechanism(s) involved it will be necessary to study spaceflight stress response at the genomic level using model organisms. Specifically we plan to: A. Expose the model bacteria Bacillus subtilis and Staphylococcus epidermidis to: (i) simulated microgravity in the laboratory and (ii) actual spaceflight in 2 parallel Biological Research in Canisters (BRIC) canisters (BRIC-A and BRIC-B) aboard the International Space Station (ISS), with appropriate ground controls. Cells cultivated in BRIC-A will be fixed rapidly with RNA Later and cells in BRIC-B will be frozen in a viable state for return to Earth. B. Total RNA will be extracted from cells in BRIC-A and subjected to transcriptome analysis using Whole Transcriptome Shotgun Sequencing (a.k.a. RNA sequencing or "RNA-Seq") at the University of Florida Interdisciplinary Center for Biotechnology Research (UF-ICBR), to identify the suite of stress responses induced by exposure to spaceflight or simulated microgravity. Cells in BRIC-B will be thawed and immediately assayed for (i) viability, (ii) resistance levels to a battery of antibiotics in the Omnilog system, and (iii) rates of spontaneous mutation to antibiotic resistance. This report documents the activities related to “Global transcriptome profiling to identify cellular stress mechanisms responsible for spaceflight-induced antibiotic resistance”. The original grant was to perform the spaceflight experiment “Biological Research in Canisters #21 (BRIC-21), with the aim of understanding how bacterial cells perceive and respond to the human spaceflight environment. We have successfully completed all the original aims and goals of this grant proposal. New Objective: In 2019, this grant was tasked with a new objective: to develop and test in spaceflight a new sample compartment for use in BRIC-PDFU and/or BRIC-LED hardware. In all previous BRIC experiments, the sample compartment consisted of the bottom half of a standard 60-mm Petri dish. However, during the BRIC-21 experiment we noted that liquid cultures assumed a toroidal shape in spaceflight (FL), in contrast to the disk-shaped ground control (GC) samples. It was hypothesized that this difference in culture geometry might account in part for the measured differences in the transcriptomic responses between FL and GC samples (Morrison, M.D., P. Fajardo-Cavazos, and W.L. Nicholson. 2019. Comparison of Bacillus subtilis transcriptome profiles from two separate missions to the International Space Station. npj Microgravity 5:1. doi: 10.1038/s41526-018-0061-0). In collaboration with the laboratories of Drs. Mark Weislogel and Titel Jurca, we have conceived and built a new sample compartment designed to mitigate the effects of microgravity on liquid culture geometry in petri dish fixation units (PDFUs). Drop-tower and biocompatibility testing of a prototype device was in progress when our labs were closed due to the COVID-19 pandemic, which essentially halted all progress on the project for several months. During the past year (10/01/2020 to 09/30/2021), we have resumed laboratory operations and have completed testing of the new modified Petri dish insert. It is now ready for utilization in spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Exposure of microorganisms to spaceflight provokes a variety of stress responses, some of which have been postulated to lead to increased antibiotic resistance. The central goal of this proposal is to identify these stress responses, using high-throughput global transcriptome and phenotypic analyses. Understanding how potentially pathogenic microbes might develop resistance to antibiotics in response to stress has significance both for long-term human space exploration and for medical science on Earth.
Task Progress:	The major factor influencing the behavior of microbes growing in liquids in space is microgravity. We recently measured the transcriptomic response of the Gram-positive bacterium Bacillus subtilis to the microgravity environment inside the ISS in spaceflight hardware called Biological Research in Canisters-Petri Dish Fixation Units (BRIC-PDFUs). In two separate experiments in the ISS, dubbed BRIC-21 and BRIC-23, we grew multiple replicates of the same B. subtilis strain in the same hardware, growth medium, and temperature with matching ground control samples (npj Microgravity 5:1. doi: 10.1038/s41526-018-0061-0). [Ed. Note: See Cumulative Bibliography listing.] In both experiments, we observed similar responses of the transcriptome to spaceflight. However, we also noted that the liquid cultures assumed a different configuration in microgravity (a toroidal shape) compared with the ground control samples (a flat disk shape), leading us to question whether the transcriptome differences we observed were a direct result of microgravity, or a secondary result of the different liquid geometries of the samples affecting, for example, oxygen availability. To mitigate the influence of microgravity on liquid geometry in BRIC canisters, we have designed an insert to replace the standard 60-mm Petri dish in BRIC-PDFU or BRIC-LED sample compartments. In this design, liquid cultures are expected to assume a more disk-like configuration regardless of gravity or its absence. We have: (i) constructed a prototype device by 3D printing; (ii) evaluated different starting materials, treatments, and coatings for their wettability (i.e., hydrophilicity) using contact angle measurements; (iii) confirmed that the device performs as designed by

	drop-tower testing and; (iv) performed material biocompatibility studies using liquid cultures of Bacillus subtilis and Staphylococcus aureus bacteria. Future microgravity testing of the device in the ISS is planned.
Bibliography Type:	Description: (Last Updated: 07/25/2025)
Abstracts for Journals and Proceedings	Nicholson WL, Fajardo-Cavazos P, Turner C, Jurca T, Weislogel M. "Design and testing of a new insert to mitigate microgravity effects on liquid cultures in BRIC-PDFU or BRIC-LED hardware." Presented at the 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5-6, 2020. Abstracts. 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5-6, 2020. , Nov-2020
Abstracts for Journals and Proceedings	Nicholson WL, Fajardo-Cavazos P, Currie TM, Jurca T, Weislogel M. "Biocompatibility testing of a new BRIC hardware insert designed to mitigate microgravity effects on liquid cultures in spaceflight." 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021. Abstracts. 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021; Abstract submission ID # 2021102. , Nov-2021
Articles in Peer-reviewed Journals	Leehan JD, Nicholson WL. "Environmental dependence of competitive fitness in rifampin-resistant rpoB mutants of Bacillus subtilis." Appl Environ Microbiol. 2022 Mar 8;88(5):e0242221.https://doi.org/10.1128/aem.02422-21 ; PMID: 35258334 PMCID: PMC8904044 ; , Mar-2022
Articles in Peer-reviewed Journals	Nicholson WL, Fajardo-Cavazos P, Turner C, Currie TM, Gregory G, Jurca T, Weislogel M. "Design and validation of a device for mitigating fluid microgravity effects in Biological Research in Canister (BRIC) spaceflight hardware." Front. Space Technol. 2021 Dec 22;2:797518. https://doi.org/10.3389/frspt.2021.797518 , Dec-2021